

BIG BAY, MI, USA

WMO: 997988

Lat: 46.8269N

Lon: 87.7266W

Elev: 185

StdP: 99.12

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.8	(c) -18.0	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 11.9	(k) -5.8	(l) 10.8	(m) -6.6	(n) 2.3	(o) 230	(p) 0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.4	(c) 29.1	(d) N/A	(e) 27.2	(f) N/A	(g) 25.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.1	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.4	(b) 8.4	(c) 7.1	(d) DB	(e) -24.6	(f) 32.4	(g) 4.9	(h) 1.1	(i) -28.1	(j) 33.1	(k) -31.0	(l) 33.8	(m) -33.7	(n) 34.4	(o) -37.3	(p) 35.2	(q) N/A
(a) 10.4	(b) 8.4	(c) 7.1	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-7.8	-7.8	-2.4	3.5	9.2	14.2	19.4	18.8	15.2	8.2	2.0	-4.1
	DBStd	10.59	5.74	5.68	6.07	4.21	4.62	3.95	3.51	3.22	4.19	4.33	5.06	4.69
	HDD10.0	2493	552	499	387	202	72	7	0	0	4	87	244	437
	HDD18.3	4727	810	733	643	446	287	133	31	31	112	315	491	696
	CDD10.0	949	0	0	3	6	46	134	290	274	160	32	4	0
	CDD18.3	142	0	0	0	0	3	11	62	47	17	2	0	0
	CDH23.3	938	0	0	5	1	71	101	367	276	106	13	0	0
	CDH26.7	212	0	0	0	0	29	28	71	59	24	2	0	0
(14) Wind	WSAvg	2.5	3.0	3.0	2.7	2.6	2.3	1.8	2.1	2.0	2.4	2.6	2.8	2.8
(15) Precipitation	PrecAvg	775	40	34	50	62	75	81	90	62	91	83	58	49
	PrecMax	1010	117	89	100	124	160	136	155	141	185	142	105	107
	PrecMin	573	15	3	3	17	30	20	39	20	35	38	27	5
	PrecStd	109	23	18	26	31	32	31	30	27	40	32	21	24
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	7.8	21.9	21.6	31.0	30.2	30.8	31.5	29.9	25.1	17.7	9.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	4.3	4.4	12.8	16.6	24.7	26.6	28.6	28.3	26.1	21.0	14.0	5.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	2.3	2.0	9.7	13.2	20.2	23.4	26.9	26.3	23.9	17.9	11.5	3.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	0.5	0.1	6.3	10.4	17.0	20.9	25.3	24.7	22.0	15.1	9.4	2.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	9.0	10.2	8.7	9.9	9.8	9.4	9.0	9.2	7.5	6.6	5.9
		MCDBR	8.8	11.1	13.6	14.4	15.0	13.4	11.2	11.0	11.6	12.5	10.6	8.0
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.269	0.286	0.314	0.343	0.379	0.382	0.389	0.372	0.363	0.332	0.301	0.267
	taud		2.440	2.377	2.372	2.379	2.362	2.388	2.403	2.448	2.468	2.518	2.521	2.491
	Ebn at Noon		839	891	913	913	886	883	871	873	850	828	793	803
	Edh at Noon		66	87	102	112	118	116	113	104	93	76	62	57
(44) All-Sky Solar Radiation	RadAvg		1.02	1.88	3.35	4.53	5.50	5.97	6.08	5.18	3.79	2.23	1.15	0.77
	RadStd		0.08	0.24	0.34	0.49	0.39	0.44	0.31	0.34	0.23	0.27	0.16	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page